



Agalic RODRIGUEZ-DUBOC

MD, PHD IN MOLECULAR AND CELLULAR NEUROBIOLOGY

Trained as a general physician and passionate about neurodevelopmental and neurodegenerative research, I focus on the molecular and cellular bases of human neuropathologies.

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🏠 Trondheim, Norway
🔗 <https://agalic-rd.com/>
🚩 French
☎ 92915242

Languages

French

Native language

Spanish

Native language

English

Native proficiency

Norwegian

Intermediate (CEFR: B1)

Italian

Beginner (CEFR: A2)

German

Beginner (CEFR: A2)

Assets

Determination

Creativity

Curiosity

Autonomy

Professional interests

Molecular biology and signaling pathways

Neurosciences, neurology, psychiatry

Preventive medicine and public health

Personal interests

Reading and literature

Painting and drawing

Knitting and crochet

Nature and sustainability

Research Experience

Since Feb. 2024

Postdoctoral Fellow

Kavli Institute for Systems Neuroscience Trondheim, Norway
"AD-CORE: Alzheimer's Disease - Cellular Origins and Reelin Effects"

From Sept. 2019 to May 2023

Doctoral Candidate

INSERM U1245 CBG Rouen, France

"Cellular, molecular and vascular alterations induced by apnea of prematurity in the murine cerebellum."

From Jan. 2019 to Jun. 2019

Master's Research Internship

InsERM U1239 DC2N Mont-Saint-Aignan, France

"Effects of perinatal hypoxia on the expression of genes involved in oxidative stress."

From Aug. 2015 to Apr. 2017

Medical Dissertation

Departamento de Investigación, PUCMM University Santiago de los Caballeros, Dominican Republic

"Psychological factors associated with nicotine addiction in college students."

From Mar. 2014 to Dec. 2014

Research Assistant Experience

Laboratory of Nanosciences, PUCMM University Santiago de los Caballeros, Dominican Republic

"Hypercaloric effect of biofunctionalized carbon nanotubes as a photothermal agent on *Staphylococcus aureus*."

Education

From Sept. 2019 to Jun. 2023

PhD in Molecular and Cellular aspects of Biology (Neuroscience)

Rouen University Mont-Saint-Aignan, France

From Sept. 2018 to Jun. 2019

Master's in Research: Neurosciences (MSc)

Rouen University Mont-Saint-Aignan, France

From Sept. 2011 to Jun. 2017

Medical Doctor (MD)

PUCMM University Santiago, Dominican Republic

Publications

- **Rodriguez-Duboc, A.**, Racine, C., Basille-Dugay, M., Vaudry, D., Gonzalez, B., Burel, D., 2026. Innovative 3D-Image Analysis of Cerebellar Vascularization Highlights Angiogenic Gene Dysregulations in a Murine Model of Apnea of Prematurity. *The Cerebellum* 25, 62. <https://doi.org/10.1007/s12311-026-02006-1>
- **Rodriguez-Duboc, A.**, Velicer, S.T., Kobro-Flatmoen, A., 2025. Bnip3 expression in the brain of an Alzheimer's disease rat model. *Brain Mech.* 148–150, 202518. <https://doi.org/10.1016/j.bramec.2025.202518>
- **Rodriguez-Duboc, A.**, Basille-Dugay, M., Debonne, A., Rivière, M.-A., Vaudry, D., Burel, D., 2023. Apnea of prematurity induces short and long-term development-related transcriptional changes in the murine cerebellum. *Curr. Res. Neurobiol.* 5, 100113. <https://doi.org/10.1016/j.crneur.2023.100113>
- Leroux, S., **Rodriguez-Duboc, A.**, Arabo, A., Basille-Dugay, M., Vaudry, D., Burel, D., 2022. Intermittent hypoxia in a mouse model of apnea of prematurity leads to a retardation of cerebellar development and long-term functional deficits. *Cell Biosci.* 12, 148. <https://doi.org/10.1186/s13578-022-00869-5>

Awards and grants

Nasjonalforeningen for folkehelsen 2023
Postdoctoral grant for dementia research

Janssen France 2022 Prize for women in science

JED EDNBISE 2022 Public prize for the best poster

Posters

ADTC AD translational conference 2025
Towards an in vitro model of early Alzheimer's disease pathology

JED EDNBISE 2022 Apnea of Prematurity induces molecular, cellular, and vascular alterations in the murine cerebellum: focusing on angiogenesis.

LARC Neurosciences 2019 Effects of perinatal hypoxia on oxidative stress-related gene expression in the murine cerebellum.

Talks

NeuroSys KISN 2025 AD-CORE: Alzheimer's Disease - Cellular Origins and Reelin Effects

ISTW Disabilities and Integration 2022
Cellular and molecular alterations induced by Apnea of Prematurity in the murine cerebellum.

JED EDNBISE 2021 Therapeutic potential of neurotrophic factors on cellular and molecular alterations induced by Apnea of Prematurity in the murine cerebellum.

Computer skills

Adobe Photoshop

Cytoscape

IMARIS

Adobe Illustrator

R

PRISM

ImageJ/Fiji

Python

Teaching and Supervision

From May 2024 to Sept. 2024

Supervision of a Master student
Kavli Institute for Systems Neuroscience Trondheim, Norway

- Supervision of a MSc/Engineering student during 4 months.

From Sept. 2019 to Mar. 2023

Supervision of Students and Interns
Rouen University Mont-Saint-Aignan, France

- Supervision of a MSc intern during 3-month
- Training of several students rotating through the laboratory

From Oct. 2014 to Nov. 2015

Teaching Assistant
PUCMM University Santiago, Dominican Republic

- Topics: Anatomy, Embryology, and Histology.
- Lab and practical sessions

Medical Experience

From Mar. 2017 to May 2017

Social and Family Medicine Internship
Centro de Salud Integral Bella Vista Dominican Republic
HIV Clinic; Epidemiology; ER

From Oct. 2016 to Feb. 2017

Surgery and Pediatrics Internships
Hospital Metropolitano de Santiago Dominican Republic
Surgery; Pathology; Oncology; Pediatric and Neonatal ICU

From Aug. 2016 to Mar. 2017

OB/GYN and Psychiatry Internships
Hospital Regional Universitario José María Cabral y Báez Santiago de los Caballeros, Dominican Republic
Peripartum; Partum; Ambulatory / Consults

From May 2016 to Aug. 2016

Internal Medicine Internship
Saint Barnabas Hospital New York, NY, United States
Neurology; Rheumatology; Pneumology; Nephrology

Practical Skills

Laboratory and Experimentation

- Preparation and mounting of anatomo-pathological samples
- Bacterial culture (agars, selective media, antibiogram)
- Preparation and analysis of blood smears
- Molecular biology (RNA extraction, reverse transcription...)
- Laser capture microdissection
- Quantitative RT-PCR
- Pipetting system robot programming
- Confocal microscopy
- Immunohistochemistry
- Transparisation and tissue clearing
- Cell culture and iPSC differentiation

Research and Communication

- Experimental design
- Statistical modeling (GLM, GLMM...)
- Scientific communication
- Theoretical and practical classes
- Writing scientific reports
- Scientific illustration
- Grant application writing
- Translation and simultaneous interpretation

Surgery and clinical skills

- Minor surgery
- Pre and post-operative follow-up
- Complete physical examination
- Differential diagnosis
- Pharmacology